Pat Oleniuk - Software Engineer

Data | AI | Full Stack | Internal Tools | SpaceX | Netflix | CERN | Former People Manager

EXPERIENCE — 10+ Years

Netflix — **1y, Remote, CA**
Senior Software Engineer - AI Observability Platform — *2026*

- Led unification of 3 data logging libs into a single solution processing **50M events/sec, resulting in ~$2M/yr savings.**
- **Saved $3M/year in compute and storage costs** for ML data pipelines by identifying and deleting unused data processing.
- Improved **training data quality** and debuggability via better schema evolution, lineage, and reduced obfuscation.
- Extended internal applications to better audit agents, MCPs, skills, features, serving etc. (see AI Lifecycle at Netflix article).

SpaceX - Starlink (Space flight and satellite internet provider) — **5y, Los Angeles, CA**
Senior Software Engineer (returned to IC to build next-gen alerting) — *2025*
*Lead Software Engineer (**promoted to people manager**; led a team of 5)* — *2022*
Software Engineer II — *2021*

- Enabled 100x customer (50k→5M+) and 10x asset (1k→30k) growth by building fleet automation and observability tools with aggregated monitoring, improved alerting, and clear UIs.
- Cut alert resolution time 5x by leading rebuild of alerting infra (collab with 40 engineers) with resilient APIs and pipelines (Java/Flink/C#/Kafka) ingesting 5M events/sec into 5 systems; widely adopted across SpaceX.
- Maintained high-availability, multi-region, **real-time data products** used by 1k+ internal users **24/7/365** to operate 7k+ satellites and other fleet/network settings. Used to: update SW, control, notify on anomalies, audit and automate actions.
- Saved **$6M+/yr** by adding a new alert schema which unlocked **machine learning** engineers to train churn-reducing models.
- Drove 2000-users-wide safety and access control overhaul following a **1hr global outage,** resulting in a system more robust to operator mistakes and **zero global outages** since then.
- Conducted **30+ on-call** incident investigations resulting in **hotfixes** preventing outages and **catastrophic space collisions**.
- Saved **$500K+/yr** by co-creating a self-service change scheduling tool automating 3 network engineers.
- Rearchitected **non-scalable** software update backend and frontend to enable daily software rollouts across 5M+ devices and 20+ concurrent A/B experiments (paper: Over-the-vacuum Software Update for Starlink).
- Boosted **productivity by 30%** by bringing 7 previously siloed teams together and reduced duplicate work.
- Grew team from 2 to 5 multi-site reports; coached via 1:1s and standups, promoted 4, and earned excellent manager score.
- Championed daily usage of our own apps (**'dogfooding'**) to quickly identify regressions and better understand feedback.

Virgin Hyperloop (aka Hyperloop One, mag-lev transportation startup, 300 people at-peak) — **5y, Los Angeles, CA**
Lead Software Engineer (IC tech-lead) — *2019*
Software Engineer — *2017*

- Reduced **time-to-insight** 50x by building a cloud simulation orchestrator and analysis pipeline for large-scale simulations and design grid search (Go, Python, PostgreSQL, Spark).
- Increased simulated transportation efficiency 70% by introducing a demand-prediction ML models (Keras, ARIMA).
- Built interactive **plots & dashboards** using Plotly, Dash, Bokeh, React allowing 4 other engineers deep-dive into data which resulted in >20 bugfixes and AI algorithms upgrades.
- Hired and mentored 3 junior developers, presented at dev conferences: Orchestrating Mass Scaled Hyperloop Simulations.

CERN (European Organization for Nuclear Research) — **2y, Geneva, Switzerland**
Test Software and Electronics Team Leader (IC tech-lead) — *2016*
Test Software and Electronics Engineer — *2015*

Full list of my experience on LinkedIn or patryk-oleniuk.github.io: Samsung, National Instruments, various ML/AI projects.

CORE SKILLS

Backend: Python, Go, C#, Java, C++, C; **APIs**: HTTP/s, Protobuf, gRPC; **Frontend**: TypeScript, Angular, React;
Deployment: Docker, Kubernetes, Bazel, Helm; **Observability & Reliability**: Grafana, Metabase, Prometheus, Influx;
Data Engineering: SQL, Postgres, Kafka, RabbitMQ, Flink, Spark, Airflow, Chronon, ElasticSearch, Cassandra, Redis;
Data Science: pandas, numpy, time series forecasting, signal processing, image processing, convolutional neural networks;
Leadership: hiring, performance tracking, radical candor, empathy, executive summary, vision, communication up/down.

EDUCATION

EPFL - École Polytechnique Fédérale de Lausanne — **2y, Lausanne, Switzerland**
Masters: Electrical Engineering and Computer Science, Focus: Data Science, Machine Learning — *2016-2018*

Warsaw University of Technology — **3y, Warsaw, Poland**
Bachelors: Electrical and Computer Engineering, Focus: Telecommunication and Multimedia — *2011-2015*